EXP 13	Case study on limitations of low-code platforms and proposing hybrid solutions

Aim:

To study the limitations of low-code platforms using the **Event Registration System** built in Zoho Creator and to propose hybrid (low-code + pro-code) solutions.

Theory:

Low-code platforms help to build apps quickly using drag-and-drop tools. But they also have limits in customization, performance, integration and cost. A **hybrid solution** uses low-code for the common parts and normal coding (pro-code) for the special parts.

Case Study: Event Registration System

The Event Registration System lets participants register for events, checks seat limits, sets the status as Confirmed or Waitlisted, sends emails and shows a dashboard. While building it, the following limitations were seen.

Limitations Observed:

1. **Limited UI design:** The look of forms can be changed only within the given options.
2. **Plan limits:** The trial plan limits the number of apps, users and permission sets. A custom permission set could not be saved.
3. **AI features not always available:** The Zia assistant did not work properly on the trial account.
4. **Builder issues:** Section fields could not be dragged into the form in some cases.
5. **Script limits:** The old value of a field (old.) is not available in the Successful form submission (Edited) workflow.
6. **Vendor lock-in:** The app and Deluge scripts work only on the Zoho platform.
7. **Performance with large data:** Complex reports may become slow with many records.
8. **Complex integrations:** Payment gateways or QR-code tickets need extra code or paid add-ons.

Proposed Hybrid Solutions:

Limitation	Hybrid Solution
Limited UI design	Build a custom front-end page using HTML/JS widgets
Plan limits	Upgrade the plan or manage roles with fewer profiles
AI not available	Call an external AI API using a custom function

Limitation	Hybrid Solution
Script limits	Store the old value in a hidden field before editing
Vendor lock-in	Export data regularly and use REST APIs
Large data	Move heavy processing to Zoho Catalyst functions
Complex integrations	Use Catalyst / NodeJs or Java functions for payments and QR codes

Observation:

Feature	Low-Code Only	Hybrid
Development speed	Very fast	Fast
Customization	Limited	High
Integration	Basic	Advanced
Skill needed	Low	Medium
Scalability	Moderate	High

Result:

The limitations of low-code platforms were studied using the Event Registration System, and hybrid solutions combining Zoho Creator with custom code and Zoho Catalyst were proposed.